



Tara Chand
Roll No.: 230104108
B.Tech - Civil Engineering
Indian Institute of Technology, Guwahati

+91-70562 02923
t.chand@iitg.ac.in
tarachandgarg79@gmail.com
tarachand.tech | GitHub | LinkedIn

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Guwahati	8.55 (Current)	2023-Present
Senior Secondary	Central Board of Secondary Education	94.4%	2022
Secondary	Board of School Education, Haryana	92.8%	2020

EXPERIENCE

- Outsyde - Adventure Booking Platform** May 2026 – Jul 2026
Full-Stack Developer Intern Remote
 - Designed a **51-table PostgreSQL database** with **Prisma**, connecting treks, trips, bookings, payments, referrals, and payouts. Used relational constraints, price and address snapshots, soft deletes, and **append-only audit logs** to maintain data accuracy.
 - Built a **9-section admin console** with role-based permissions across **64+ APIs**, covering provider KYC, user management, refunds, disputes, commissions, and payouts. Fixed a payment race condition using **row-level locking** to prevent duplicate refunds.
 - Developed the booking platform with consumer and provider portals, **retry-safe payments** with duplicate-charge protection, and a loyalty wallet system. Added discovery features such as fuzzy search, trending rankings, and sponsored listings for growth.

PROJECTS

- GameHub - Multi-Game Platform** | *Group Project* Mar 2026 - Apr 2026 | Website
 - Built a multi-game platform using Node.js, Express, and MongoDB, with a shared game engine handling scoring and game rules across **5+ games**, providing consistent structure and reusable game logic while allowing new games without backend changes.
 - Implemented **JWT + bcrypt** authentication with **OTP email verification** for signup and password reset, using Brevo for emails.
 - Built **real-time multiplayer** for Ludo using **Socket.IO**, with server-side turn and dice validation to prevent game-state manipulation.
- Smart Traffic Command Center** | *Group Project* May 2026 - Jun 2026 | Website
 - Engineered **64 predictive features** from **8,173** historical traffic incidents and trained chained **gradient-boosting** models, achieving **ROC-AUC 0.989** on closure classification using a leakage-safe pipeline with clusters, priority encodings, scalars fit on training data.
 - Built a data-driven severity score using **PCA-weighted blending** across 7 indicators, generating operational recommendations.
 - Built a **Streamlit** dashboard with an interactive map predictor, model thresholds, and prediction history using crash-proof writes.
- TextLite - Terminal Text Editor** | *Personal Project* Jul 2026 | Github
 - Built a terminal text editor in **C++17** using a custom singly linked list for in-memory line storage, supporting create, edit, search, delete, and compile operations through a menu-driven CLI with zero external libraries, syntax highlighting, and file management.
 - Designed an encapsulated **TextEditor** class, eliminating global state with explicit manual memory management using raw pointers.
 - Implemented **O(1)** line insertion and deletion through pointer-based linked-list splicing, validated with unit tests for boundary cases.
- MathSage - RAG CLI for Math Q&A** | *Personal Project* Jul 2026 | Github
 - Developed a **Retrieval-Augmented Generation** pipeline in **Python** using **LangChain**, **Chroma**, and **Gemini**, preserving **multi-turn conversational context** while grounding responses in live Stack Exchange data using vector similarity search.
 - Indexed **30+ documents** per query with **sentence-transformer** embeddings, retrieving the **top-3** candidates for generation.
 - Designed an **LCEL retrieval chain** (retriever, prompt, LLM, parser) for swappable stages and prompt-constrained generation.

TECHNICAL SKILLS

- Languages:** C, C++, Python, JavaScript, TypeScript
- Web & Backend:** React, Next.js, Redux, Tailwind CSS, Node.js, Express, REST APIs, WebSockets/Socket.IO, JWT/OAuth
- Databases & Services:** MySQL, PostgreSQL, MongoDB, Prisma ORM, Firebase
- Data Science & ML:** NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, TensorFlow, Keras, LangChain
- Core Fundamentals:** DSA, Object Oriented Programming, DBMS, OS, Probability, Statistics
- Tools & Other:** Git, GitHub, Postman, Docker*, MATLAB*, L^AT_EX* * Elementary proficiency

ACHIEVEMENTS

- Top 75 nationally**, among **30K+** registrations, selected for the final code-a-thon at Cisco campus, Bangalore 2026
- Semifinalist**, Flipkart GRiD 8.0, India's largest engineering challenge, Software Development Track 2026
- Semifinalist**, Flipkart Gridlock 2.0, qualified with the Smart Traffic Command Center project 2026
- Semifinalist**, among **47K+** registrations, MyCareernet HackVega 2.0, national software hackathon 2026
- Specialist** at Codeforces, Global Rank **759** among **20K+** participants in Round 1117 (Div. 2) Link
- 3★** at CodeChef, (Max. Rating **1650**) Global Rank **146** among **5K+** participants (Starters 236) Link
- Competitive Programming**, Solved **1100+** DSA problems across Codeforces, CodeChef, LeetCode & AtCoder Card

EXTRA-CURRICULAR ACTIVITIES

- Student Mentor**, Mentored JEE aspirants at Physics Wallah 2024
- Participant**, Meta Hacker Cup, Meta's global competitive programming contest 2024
- Participant**, Mime Battle, Manthan (Inter-Hostel Cultural Competition), IIT Guwahati 2024
- Core Team**, Public Relations and Branding, Alcheringa'26 (Cultural Fest), IIT Guwahati 2026